

ENTRY 031 — The Model Doesn't Know When Entanglement Actually Happens — August 13, 2026 — Status: COMPLETE

Entry 031 — Route-Then-Entangle: A Structural Assumption Error

The Outlier Edge Was a Red Herring; the Real Bug Is About WHEN a SWAP Chain Actually Starts Carrying Entangled Information

1. Ruled Out: The Closed Form Doesn't Break at High p

Entry 030 flagged Sherbrooke edge (57,58) -- gate_error 11.68%, ~15x the chip median -- as a possible case where the closed-form baseline (verified on Kyiv up to ~4.3% error) simply doesn't hold at larger magnitudes. Isolating that exact edge alone, sweeping gate count the same way Entry 025 did, found the closed form matches Aer to within 0.13 points on average even at p=11.68% -- essentially exact. The formula itself is not the problem, at any error magnitude tested so far.

2. The Real Bug

Re-isolating the FULL bell_near_57_61 route (4 real edges, matching the exact transpiled circuit used for the Aer reference) reproduced Entry 030's gap directly: 98.71% real gate-error-only survival against 83.5% predicted. Since the per-edge formula is now proven correct, the bug had to be in how results from different edges combine. A minimal toy circuit isolated it: build a Bell pair by applying H to one qubit, then routing BOTH halves toward each other via SWAPs, and creating the actual entanglement with a SINGLE cx/ecr gate only once they're adjacent -- exactly what the real transpiler does for these circuits, confirmed by inspecting the transpiled gate sequence directly (only one non-SWAP two-qubit gate appears, at the very end).

Every model since Entry 023 assumed the opposite: that entanglement exists from the start and every SWAP afterward is moving an already-entangled payload, equally vulnerable to depolarizing damage at every hop. That assumption is wrong for exactly the circuits this project measures. Before the final entangling gate, the two qubits being routed toward each other are carrying ordinary, UNENTANGLED product-state information -- and a depolarizing event on a product state does not destroy a correlation that doesn't exist yet. It does less damage to the final {00,11} outcome than the same event would do to an already-entangled pair.

test	Aer	closed-form (old model)	diff
toy: entangle-first-then-route (wrong structure)	50.32%	93.62%	-43.30
toy: route-first-then-entangle (real structure)	99.29%	93.62%	+5.67
real bell_near_57_61, gate-error-only	98.71%	83.52%	+15.19

The first row is a deliberately broken toy (re-entangling an already-entangled pair) included only to show how sensitive this structure is -- getting the entanglement-timing wrong doesn't cause a small error, it inverts the whole prediction. The second row, with the structure corrected to match what Qiskit's transpiler actually produces, shows the SAME direction and rough scale of under-prediction as the real circuit in row three. That consistency is what makes this the real explanation rather than a coincidence.

Entry 031 — Conclusion

The persistent over-prediction of damage on long, multi-hop routes (bell_mid and bell_far on both Kyiv and Sherbrooke, and now bell_near_57_61) traces to one structural assumption: exact_dwell_cost and route_cost both treat every SWAP identically, as if it always carries an already-entangled qubit. Real transpiled circuits for a single Bell/GHZ preparation typically route the uninvolved qubits into position FIRST and entangle them LAST, meaning most of a long route's SWAP legs are moving product-state information that is measurably less vulnerable to depolarizing noise than the model assumes. This is not a tuning problem or an edge-magnitude problem -- it is a category error in what physical process the model thinks is happening during most of a circuit's real gate sequence.

3. Why This Wasn't Caught Earlier

Entry 025's isolation test, which the closed form was derived and verified against, deliberately used "cancelling CX pairs" applied repeatedly to an circuit that is ALREADY entangled from the first gate -- a clean, controlled test of the

compounding formula, but one that never exercises the pre-entanglement regime a real multi-hop route spends most of its gates in. The formula was proven correct in exactly the case it was tested in, and that case turns out not to be representative of the majority of gates in a long real route.

4. Next Step — Entry 032

This entry identifies the mechanism but does not yet fix it -- deriving the correct closed form for a depolarizing event on a pre-entanglement product-state SWAP (as opposed to a post-entanglement one) needs its own Kraus-operator derivation, analogous to Entry 025's and Entry 029's, before it can be verified and shipped. The practical implication for `exact_dwell_cost`: gate error needs to be split into two categories per route -- SWAPs before the pending pair's entangling gate, and SWAPs/gates after it -- rather than one uniform survival product across the whole route.

Research Log — Index Addendum (Entry 031)

Entry	Title	Date	Status
Entry 031	Route-Then-Entangle: A Structural Assumption Error in Gate-Error Modeling	Aug 13, 2026	Complete (root cause found, fix pending)

entry031_high_p_isolation.py confirms the closed-form baseline holds at p=11.68% (rules out edge-magnitude as the cause). entry031_route_then_entangle.py isolates and confirms the real cause. Neither emulator_v4.py nor exact_dwell_routing.py is fixed yet -- left for Entry 032.